#8 · 5.9% of marks

Ch 12 — Interest Rate Risk Management

Numerical every attempt, different tool each time: FRA (Sep 2025), cap & floor (Jan 2026), gap / Earnings at Risk + zero-cost collar (May 2026). SM TYK theory Qs: **Forward Rate Agreements** · **swaptions and their uses**.

① Interest rates & benchmarks (SM 1–2)

→ *Interest = cost of borrowing + compensation for service and **risk** of lending (default) + protection against **inflation**.*
__Factors__ (macro): (a) demand / supply of money (b) inflation (c) government — biggest borrower; RBI via note issue / __OMO__ changes CRR, SLR, bank rate

→ ***Benchmark (reference) rate**: basis for other rates — loans, overdrafts, mortgages, forwards, futures, options and especially **swaps**; floating loans = benchmark + bps by credit rating; decided by an independent body; also used to value bonds*

→ ***LIBOR**: manipulation news 2012 → decided 2017 to end → from **1 Jan 2022** use **Alternative Reference Rates (ARRs)**. ARRs differ: (i) based on **actual overnight transactions** (secured or unsecured), LIBOR unsecured and relied on panel banks' judgment (ii) **near risk-free, no term premium***

Region	ARR	Regulator	Nature
USA	**SOFR** — Secured Overnight Financing Rate	Federal Reserve Bank of New York	Secured
UK	**SONIA** — Sterling Overnight Index Average	Bank of England	Unsecured
Europe	**€STER** — Euro Short-Term Rate	European Central Bank	Unsecured
Japan	**TONAR** — Tokyo Overnight Average Rate	Bank of Japan	Unsecured
Switzerland	**SARON** — Swiss Average Rate Overnight	SIX Swiss Exchange	Secured

***India**: repo, PLR, MCLR, but most common are **MIBOR** (rate bank charges a borrower) and **MIBID** (rate at which bank would borrow from another bank) — used in IRS, FRA, floating-rate debentures.*

Secured ARRs = SOFR and SARON (both start with S). The other three are unsecured.

② Interest rate risk (SM 3)

*Change in bond prices from rate changes + impact on interest income → **price and reinvestment** risks both matter.*

THEORY + GAP NUMERICAL

③ Types of interest rate risk (SM 3.1)

Type	SM meaning
Gap / mismatch	Assets, liabilities, off-BS items with different principal, maturity or **re-pricing dates**. **Positive gap** = RSA > RSL → rate **rise raises NII**. **Negative gap** = RSL > RSA → NII rises when rates **fall**. **Earnings at Risk (EaR) = gap × assumed rate change**; bank fixes an EaR limit (based on last / current income) and a trigger for on- / off-BS hedging.
Basis	Rates on different assets / liabilities change by different magnitudes (eg assets fixed, liabilities floating); high for banks funding composite assets from composite liabilities; NII expands = favourable basis shift
Embedded option	Prepayment of loans, call / put on bonds, premature withdrawal of deposits; faster and bigger rate moves → bigger risk; adjust gap statements, charge penalties based on opportunity cost
Yield curve	Yield-curve movements over business cycles affect portfolio values and income
Price	Assets sold before maturity; prices and yields inverse; tied to **trading book** → limit size, holding period, duration, defeasance period, stop-loss, MTM
Reinvestment	Uncertainty about rate at which future cash flows can be reinvested; mismatches vary NII

Gap report — 5 limitations (SM)

(1) only timing difference of re-pricing; ignores **basis and embedded option** risk (2) assumes all items in a time band re-price together; ignores effect on **economic / market value** (3) ignores timing differences in payments caused by rate changes (4) assumes **parallel shift** of yield curve — seldom happens (5) ignores variability in **non-interest** revenue & expenses

④ Measuring interest rate risk (SM 3.2)

- → Must identify and quantify first; system should cover gap, basis, embedded option, yield curve, price, reinvestment and net interest position risks
- → Techniques: **maturity gap analysis** (sensitivity of **earnings**) · **duration** (sensitivity of **capital**) · **simulation** · **VaR** — often used in combination / hybrid

May 2026 Q5(b) (ICAI): re-pricing gap = RSA − RSL (negative); EaR = gap × rate change × ¼ (quarter) vs limit of 8% of NII → hedge if EaR exceeds the limit. Rates up hurts a negative gap; rates down helps.

5 Hedging methods (SM 4)

Traditional: (i) Asset Liability Management (ALM) (ii) Forward Rate Agreement (FRA). **Modern**: (i) Interest Rate Futures (ii) Interest Rate Options (iii) Interest Rate Swaps (+ swaptions).

6 ALM (SM 4.1.1)

Evolved by **RBI**; introduced in Indian banks w.e.f. **1 April 1999**. Comprehensive, dynamic framework to measure, monitor and manage **liquidity, interest rate, forex, equity and commodity price** risks, integrated with business strategy; structure the balance sheet so **net interest earnings are maximised** within risk preference. Extends to liquidity, market and trading risk, funding & capital planning, profit planning, growth projection. First step in long-term strategic planning — an **intermediate-term planning** function.

7 Forward Rate Agreement (SM 4.1.2) — TYK Q1

- → Borrower / lender protects against adverse rate changes; **OTC** (not exchange-traded); used by banks to fix cost of future deposits / revenue of variable loans indexed to SOFR, €STER, MIBOR
- → **Off-balance-sheet**; **no transfer of principal** (notional); **cash-settled** for the difference
- → RR > FR → **seller pays buyer** (buyer = borrower protected); RR < FR → buyer pays seller
- → Settled at the **start** of the period → difference **discounted at the actual (RR) rate**

Settlement = N × (RR − FR) × (dtm/DY) ÷ [1 + RR × (dtm/DY)] (DY = 360 or 365)

SM eg: **1 × 4** FRA (deposit placed after 1 month, matures 4 months from contract) at 5%, $1 m, 90 days; SOFR 6% → 1 bp = $25 → $2,500 ÷ [1 + 0.06 × 90/360] = **$2,463.05** to buyer.

TYK 1 (app-computed): ₹60 cr, 3 months in 6 months, FRA 9.30% → at 9.60% company **receives ₹4,39,453**; at 8.80% company **pays ₹7,33,855**. TYK 2: 6 × 9 FRA ₹100 cr at 5.25%, 3-month MIBOR 5.50% → **+25 bp**, settlement ≈ **₹6,16,523**.

8 Interest rate futures (SM 4.2.1)

- → Futures on an **interest-bearing asset** — bonds are the underlying, not the rate. Borrower fearing a **rise sells** futures (prices fall when rates rise → profit on buying back)
- → NSE: futures on **6, 10 and 13-year GoI securities** and **91-day T-bill**. Settlement: **daily MTM** + **physical delivery** any day in expiry month; final settlement only on expiry date
- → **Conversion factor** (published by NSE; introduced by RBI): CF × futures price = delivery price of a given bond
- → **Cheapest to deliver**: seller's profit = futures price × CF − quoted spot price; CTD = bond that **maximises profit / minimises loss**

NUMERICAL CORE

⑨ Interest rate options (SM 4.2.2)

Also called **Interest Rate Guarantee (IRG)**: a right, not obligation — insurance against adverse moves while keeping favourable ones. An IRO is basically a **series of FRAs** exercisable at set rates each period.

Option	Who / what	Payoff (SM)
Cap (call)	Buyer (**borrower**) pays premium; receives difference when benchmark > cap rate; no obligation if below. Evolved from IRGs on floating-rate loans (1985); LBO boom — lenders required caps. Features: notional, benchmark, cap rate (= strike), period with reset / payment dates	**$N \times \max(0, r - r_c) \times d_t/\text{days}$**
Floor (put)	Buyer (**lender / depositor**) protected from **fall** in rates; = right to **sell an FRA** → European put on an FRA; multi-period floor = bundle of puts on a sequence of FRAs	**$N \times \max(0, r_f - r) \times d_t/\text{days}$**
Collar	**Buy a cap + sell a floor** → rate bounded on both sides; cheaper than a cap alone (floor premium offsets); **zero-cost collar** = floor premium exactly offsets cap premium; gives up benefit of a big fall	**$N \times [\max(0, r - r_c) - \max(0, r_f - r)] \times d_t/\text{days}$**

SM cap eg: 1-yr cap, $1 m, 6-month SOFR cap 5%, reset 15 July (first period excluded — known at inception); SOFR 5.5% → paid next 15 Jan (184 days): 1,000,000 × 0.005 × 184/360 = **$2,555.56**.

⑩ ICAI patterns

- **Jan 2026 Q4(a)**: borrower **buys cap**, depositor **buys floor**. Lump-sum premium converted to per-period: **$P = A \times r_p \div \text{PVAF}(i, n)$** ($i$ = fixed rate per period). AB Ltd ₹5 m, 0.4%, PVAF(4%, 10) 8.1108 → **₹2,466**; gain 2% × ½ × 5 m = 50,000 → net **₹47,534**. XY Ltd ₹2 m, 0.5%, PVAF(4%, 6) 5.2421 → **₹1,908**; net **₹18,092**. ICAI also accepted $n - 1$ resets.
- **Cap to peg at 9%** (MTP): pay actual rate, receive excess over 9% → net cost never above 9%; years at or below 9% → option lapses. Cap preferred to a collar here — selling the floor at 9% causes outflow when rates fall below 9%.
- **May 2026 Q4(c)**: zero-cost collar — value of cap = value of floor (PV of expected payoffs) → solve the floor rate.

> ⚠ **TRAP**
> Cap / floor is per **reset period** — multiply by the fraction of the year (½ for 6-month resets). The first period is usually excluded (rate already known).

11 Interest rate swaps (SM 4.2.3)

- → Counterparties exchange payments indexed to two different rates on a **notional** principal that is **never exchanged**. Early swap dealers took a brokerage fee only; contract was between the end users
- → **Fixed / floating swap**: fixed rate · floating rate reset periodically · notional · term with reset and payment dates. Settlement **2 days after trade date**. Fixed rate quoted as **All-In-Cost (AIC)** against a flat floating index
- → Fixed payment (semi-annual) = $N \times AIC \times 180/360$ (30-day months, 360-day year); floating receipt = $N \times R \times d_t/360$

Type	SM meaning
Plain vanilla (generic / coupon)	Fixed ↔ floating on notional; floating = SOFR, €STER, SONIA, MIBOR, PLR
Basis rate (non-generic)	Both legs floating on different benchmarks (eg 1-month vs 3-month SOFR)
Asset swap	Fixed-rate investments (bonds) swapped for floating-rate investments (an index)
Amortising	Notional **declines** over the life — for issuers of redeemable bonds / debentures (hedges the redemption profile)

12 Swaptions (SM 4.2.4) — TYK Q2

- → Option on an IRS — right, not obligation, to enter a swap at a future date, fixed rate and term. **Payer (call) swaption**: pay fixed, receive floating. **Receiver (put) swaption**: receive fixed, pay floating
- → **Features**: option on a **forward-start IRS** with terms fixed at inception · "**3-month into 5-year**" = option to enter a 5-yr IRS in 3 months · option period = trade date to expiry · premium in **basis points** · can be **cash-settled** (MTM off the forward curve). Pricing: model of distribution of the forward zero-coupon curve
- → **Uses**: (a) traders & treasurers (b) speculate / hedge swap books (c) hedge **embedded optionality** (d) borrowers targeting an acceptable rate (e) businesses **tendering for contracts** (f) protection on **callable / puttable bonds**

13 Correlation — how Ch 12 actually connects

Ch 12 idea	The actual link (app-written cross-reference; the Ch 12 side is SM)
Price & reinvestment risk	**Ch 5**: same two risks that **immunization** offsets (duration = holding period); duration measures sensitivity of capital, gap measures sensitivity of earnings.
Cap, floor, collar	**Ch 9**: cap = series of calls, floor = series of puts on the rate; collar = long call + short put — same as a zero-cost option collar on shares.
FRA settlement discounted at RR	**Ch 10**: same idea as early settlement of a forward — money paid at the start of the period is worth less, so discount at the actual rate.
IRS / swaptions	**Ch 10**: currency swaps use the same comparative-advantage logic; **Ch 6**: IRS changes the duration of a bond portfolio (active fixed-income strategy).
IRF conversion factor, CTD	**Ch 5**: bond price = PV of coupons + redemption — CF makes bonds with different coupons / maturities comparable.
SOFR / SONIA / MIBOR	**Ch 11**: syndicated loans are priced at SOFR + spread; **FR Ind AS 109**: IBOR reform and hedge accounting.